https://www.youtube.com/watch?v=gD84WMLmQes

hi in this video i will be introducing
kuhn
i will be talking about what question
he's asking and what philosophy of
science
problem he's identifying so what
characterizes his approach is that he
was a historian of science
so he looked at the history of science
in order to answer some key questions
such as
how do scientific theories actually
develop over time do they
improve cumulatively do they sort of
build on top of each other in practice
and is scientific truth or objectivity
ever completely achieved so again he
was trying to answer those questions
by looking at the actual history of
science
in that sense we can contrast it to who
has more philosophical perspective
who's just sort of thinking more out of
principle and considering
what science is and what can it do
considering the challenge it faces
so this is the then also a
sociological
perspective Kuhn is taking because he's
looking at the
practice of collectives of researchers
who
together and in competition and
cooperation try to improve on science
how does that actually happen
and these are then some of the key
concepts that we'll get a grasp of
throughout the next
videos and live session
this is just to highlight that he
Kuhn, this Kuhn text that we're reading
and extract from a chapter from
this is really one of the most cited

social science research of all time
of all social sciences so these numbers
are from 2016 so
they're a little bit old but you might
recognize the name porter
and karl marx and then maslow and and
some other
maybe you know some of the other people
on there
but this is just really to highlight
that it's not just in business
administration or in
sort of a particular subset of the
social sciences or even the general
sciences that this is interesting this is
something that goes across and
abroad a range of different fields and
has been of
of interest to these many different
disciplines

Scientific Paradigms and Disagreements

key question that he's really trying to
get to is how does science and
scientific explanation actually evolve
and a metaphor we can consider here is
that
either its science and scientific
theories are accumulating like when you
build a lego tower
so this is then first theory and then
you build another one and then on top of
this one you build the next etcetera et
cetera and in the end you might have a
big beautiful theoretical explanation of
how the world actually works
brick by brick in a very linear fashion
that's one perspective on how
science develops over time kuhn's
book which is called the structure of
scientific revolutions which then gives
an indication of that
he has a different take, so he focuses on
the revolutions that happens from time
to time
that sometimes the tower is destroyed, so
the lego tower that we're building, and

we have to start from scratch
so science is not at the position that
kuhn is arguing for
science is not one long accumulation of
just building on top of formal knowledge
and this then also relates to the
question is there actually an external
objective truth that we're trying to
capture is there
something out there that we can then
imitate and copy and
we're just trying to uncover or is there
something else going on
so it's a meter level so the abstract
perspective on what
theories are what sciences and and what
we're actually doing
in research and just again to highlight
and to emphasize the position that he's
then arguing against we have to remember
that this book was written in 1962
it's quite a while ago and a general,
sort of a little bit simplified, but a
general textbook conception before Kuhn
this was literally what was written in
textbooks
was that theories were directly compared
to the world through observations
and there are facts so to speak that we
can that we can rely on the
philosophy of science book chapter
that's also part of the curriculum also
talks about this on page 80.
so theory corresponds to the world
that's the also an essence of positivism
that Guba talks about
so when a theory is true this is then a
direct reflection of how stuff actually
works in the world
and an implication of this then is also
then theories accumulate because if the
theory you're building on is true well
then the next one has to
almost has to take that has to build on
that
and theories are tested deductively
that's the approach

um that characterizes science so this
was quite literally
the textbook conception that Kuhn wasn't
trying to argue against
based on his read of how different
sciences, different scientific fields
have developed over time
so he's arguing against this linear lego
metaphor where
scientific development is just rational
and cumulative brick by brick
and a way of trying to make sense of
this or to try to illustrate Kuhn's
skepticism is to consider this question
well if a better theory is presented
well
it should be accepted right? and this is
positivism's take
on science that it's accumulative
true theories represent how the world
actually works so if a better theory is
presented
we are in a rational scientific
community we should accept this
right?!

Rejection of Better Explanations (handwashing example)

and a counter example to this is
quite famous and in this year maybe also
fitting that it has something to do with
hand washing
so in the in the middle of the 19th
century there was a
doctor called Semmelweiswho was in
charge of a maternity ward
where women go to give birth or
that's what they did there. they went
there to give birth and get help from
from doctors like Semmelweis and what
we see is here on the y-axis we have the
monthly mortality rate so this is the
number of
mothers who died due to purple fever
after a while they were giving birth and
as you can see in some of these months
this was really quite

terrible and terrifying absolutely
terrifying numbers
a very large proportion of
mothers died in in this process i mean
also in the 40s there were also
months where it really was fantastic
but generally speaking the average
really is
quite bad and then in 47 Semmelweis
makes a change in his in his unit
he introduces hand washing so whenever a
doctor has just
dealt with a corpse which or, actually
often, was the problem they went straight
from dealing with someone who just died
to trying to
help deliver a baby, whenever they had to
get
in close contact with a mother and be
involved in a birth
they had to wash their hands very very
simple suggestion that was being made
there but remember this was the 19th
century
and people just weren't aware of this it
this is not a randomized control trial
we have to remember because this is the
same unit where he makes this change
but it's still quite noticeable that
after he introduces hand-washing we have
a lot of months
what is it here 15 or something that are
pictured here
where the mortality rate is a lot lower
these are big big
differences that we are talking about
but and this is then the key
thing that that Kuhn would then sort
of you could use this example to say
well this was there was clearly
something going on here which was
effective
we are we talking about a construct that
is very easy to define
either you are alive or you're dead it
turns out by introducing hand washing we
could reduce

the mortality rate substantially but
other doctors
in his local community and also sort of
one on a wider national and global basis
did not accept his explanation they were
sure that something else had happened
maybe Semmelweis had just become a
better doctor or maybe he was getting
different patients or
whatever other sort of confounding
reasons one could imagine
so this is a situation where there was
and then science reveals plenty of these
examples
where one could deliver a clearly better
explanation
but it wasn't immediately accepted by
the scientific community
a key reason here was probably that he
had no theory to explain this he
literally had no clue about why this was
the case
he just could you could just see that if
we wash our hands
we get um we see fewer deaths
um there are probably a couple of
reasons for why there was this
resistance and
one element is simple confirmation bias
this did not fit in the world view of
the doctors
a related element is probably also that
he's clearly putting the blame
on the doctors who are not able to do
something as simple as wash their hands
so
taking all that responsibility and being
responsible for all these mortalities
that we see before
the introduction of hand-washing is of
course also quite
devastating to have to acknowledge as a
doctor so there are probably multiple
reasons but
but in any case the clear and better
explanation is not accepted
and a challenge was probably that he did
not deliver a theory

actually if we go back to even in roman times
before Christ there was a roman statesman who
said there is some “minute creatures which cannot be seen by the eyes which can
enter the body through the mouth and then can cause serious disease
this was the first time as far as we know anyone ever talked about
germs as being a thing this was really the challenge here Semmelweis did
not know that germs could be these small little
particles that can crawl for instance from um a dead body to the doctor's hand and
then contaminate whatever else and get in touch with that simply wasn't
and a terminology there was no awareness of disease
being such a thing that could be on a hand so
there wasn't a paradigm that's what the word Kuhn would use
there where there was no sort of basic assumption then and a basic
understanding of how the world works which means a paradigm
that could help make sense of this which is then a key reason for why
this explanation was rejected for years and years
until someone called Pasteur came along and introduced and
framed the germ theory and just within a couple of decades or two to three decades
hand-washing was a thing but it's still an explanation of why
it's still an example of
how better explanations are not always accepted
many other examples could be given but of course this is sort of quite
drastic so it seems to be the case and

Acceptance of Theories and the Paradigm Shift

that's at least also the argument that
kuhn isn't making that it doesn't seem
to be the case that
just because you have a more true
explanation or you have stronger
empirical data which Semmelweis has
clearly had or he also had a very simple
explanation
it's often also not enough that the
majority agrees i mean it's not
possible when we look at the history of
science to say well it just needs to be
one of
all of these factors and then a new
theory is accepted it's a much more
messy
sociological psychological process and
that's really the key thing that we have
to understand here is so
this is how Kuhn is trying to explain
it, how
signs and theories develop all the time
it's not just about whether
it sort of captures how the world
actually works in a cumulative way
we have to understand human psychology
and human
sociology and that's what we'll be
getting an understanding of
so this is then his take on scientific
development that we are aiming to
understand
that yes very often science is
cumulative
and normal and we are building bricks on
top of each other but sometimes we are
facing an absolute crisis
and whether it's realizing that germs is
a thing
or the earth revolving around the sun
rather than the other way around we have
to start completely over and
change our paradigm and change the way
that we perceive and think of the world
so that's also sort of how we define a
paradigm it is about these fundamental
assumptions

assumptions that members of a scientific
community have we'll get
the next video will explicitly deal with
the content of paradigm
this is i'm still trying to um
present and explain the why it is
that better theories are not always
accepted
and especially the psychological angle
of it
just to make clear i mean often science
is accumulative often it is deductive
um but uh most of the time he
conceived this is sort of how stuff
goes on but from time to time we have
these sort of radical paradigmatic
shifts where we have to completely
reorient the way that we try to make
sense of the world
and this is where we sort of have a
revolution where we have to start over
again we'll get more into what a
paradigm is and the paradigm changes now

Individual differences: Perceptions and Biases

i want to
dive into the psychological process of
why is it that we might
reject a new theory and this speaks
quite directly to what Kahneman was
talking about as well
it's something about how we human
beings perceive the world it's something
about the biases that
that sort of
also shapes the way our cognition works
and it's something about sociological
norms which we'll get to as well
so a key
concept in a key aim by Kuhn is to
emphasize that we quite
literally do not see the same when we
see the world
data or the world does not prescribe
just one interpretation
we um i mean we can look at this picture

and then see all the different dolphins
that are on this picture and i don't
know whether
11 or something of them but we can also
look at the picture and then maybe see
something else and i'm sure you've seen
many of those visual illusions you can
pause the video if you want to sort of
find out where the dolphins actually are
whatever else is on this video
Kuhn's point is we don't see the world
we see a processed version of the world
in the metaphorical to quote him "in
the metaphorical no less than in the
literal
use of "seeing",
interpretation begins where perception
ends… what perception leaves for
interpretation to complete
depends drastically on the nature and
amount of prior experience and training"
so how we perceive and interpret the
world
drastically depends on the nature and
amount of prior experience and
training very much in
line with what kahneman was also talking
about
so i might have a specific scientific
sort of training and experience that
enables me to see something as
this new scientific fact but others
might not have that experience or have a
different experience and then see
something else
how do we then decide who is true or not
so this is in his point that we can't
just easily shift between these
different ways of seeing the world
between these different paradigms and in
the end theories and different paradigms
are incommersurable they are
incomparable they you don't have one
single standard that enables you to
compare them
we'll also get more back to what that
exactly means
um so this is just sort of to
highlight the same point we can look at
this and then say

is it a duck or is it a rabbit and well
we can ask
google's algorithm google has an AI
algorithm that can categorize what an
image is and as you can see
depending on how we rotate the image it
actually says
it's a duck or it says it's a rabbit so
this is just quite literal
just we run it again here quite literal
example of how
what we see depends on the angle from
which we see it from here the
object itself is rotating but in the end
it could also be
the sort of the human perspective that
is rotating depending on the
prior amount of training experience we
have so
again the point is that we literally do
not see the same
"if two people stand in the same place
and gaze in the same direction,
we must… conclude that they receive
close to similar stimuli" they see
close to the same thing but people do
not see stimuli so it's not just
about us sort of having
our heads tilted in a slightly different
angle they have sensations
and we can then interpret these
sensations in two different ways um
and then it's because the sensations will
not
per say be the same thing much neural
processing takes place between the
receipt of stimulus and the awareness of
a sensation
and the and whatever we perceive in the
world is in part conditioned by
education so whatever education and
prior training we have it
helps shape to see now this is just a
simple example whether something is a
rabbit a duck
but imagine we're looking at scientific
data

or phenomena out in the world the point
here would be to say
we can't just say well this is what
everyone should see
irrespective of their training because
Kuhn's point is that
whatever training one has conditions
ones to see things in different way
of course he's also saying you can't
make up any kind of interpretation “if
you can't distinguish a wolf from a dog
,you won't survive”
sorry from a dog so i mean it's not
saying that anything goes
but in situations where one might be uncertain he's saying
there is no clear absolute standard to refer to to know
what it is we're looking at notice also
we make an observation we don't have it
as such
and then he's he's again he's also going
to physics to make some illustrations
for instance here
he's saying we do not see electrons but
rather that tracks or else bubbles of
vapor in a cloud chamber this is
actually then again what we're looking
at here we're seeing
an electron somewhere out here and then
we're seeing some other particles in in
the kind of image or whatever we want to
call it that we're seeing here
and yeah einstein and bohr they've
actually disagreed on what they saw but
but they would probably be better at
understanding what's going on than the
me the confused person illustrated here
in the middle
because i sort of have a sense of it but
i don't have the training education
to really understand what's going on in
this picture
we don't see the electrons as such we
need to sort of
interpret and translate whatever we see
into constructs that can make sense to

us constructs that
we've gained from education and training
and different educations and different
trainings might lead us to create
slightly different kinds of constructs
whether it's
physics or it's unemployment or
innovation or strategy or whatever
concept we are dealing with so again
“in the metaphorical no less than the
literal use of seeing
interpretation begins with perception
ends” what we perceive what we see in
the world depends
drastically on the nature and amount of
prior experience and training
what we see depends in part on what we
believe and facts are never just facts
fact they're embedded within a paradigm
so this is not
a fact of an electron
circling around a particle it also starts
with
some paradigmatic assumptions about
what we see in the world
and that's also indicated just before we
don't see motivation and the
efficient market hypothesis productivity
firm performance
unemployment etc. um but we
have some assumptions about how we can
get out into the world measure things
define things
and how we can claim that they're
relations between things
um but again prior training and
experience
has enabled us is to do this is Kuhn's
argument
this is um trying to combine Kuhn and
Kahneman's
language um and an example where
someone has exposed economic researchers
to some claims and they've either said
this is a statement made by mainstream
economist or some
sort of a bit more unorthodox a bit more
sort of

outlier strange thinker and it it's it
and it turns out that if if it's not
attributed to sort of a mainstream
well-known economist
well then economists agree with it less
so the same statement will have
will be perceived to have a different
truth value depending on who the sender
is which again shouldn't be the case if
science was just something
abstract and true and corresponding to
how the world really works there are
these are human beings that have biases
that influence the way that they see
and interpret scientific findings um
this is Kuhn's point and this is in in
that sense also the same point that kind
of man was making
a couple of decades later the biases we
have
influences what we see how we filter the
world how we remember things
and how we what inferences we draw
i really like this example here that
also illustrates the same point
i would claim that if one hadn't seen
this before
one would not know what we're looking at
here
i can replay the video
i have a better sense of what we're
looking at because i've been
told and i've read about it that this is
a picture, a video so to speak of a
alien of a very distant star system
where there's a star in the middle
some so to speak that is being circled
by seven different planets this is only
a subsection of the video so we can't
see the entire thing but still
data is not raw data is not just
data we always interpret it on a
paradigmatic background and whether
we're looking at planets or whether
we're looking at a statistical model
it's never raw as such we don't see the
world as it is in itself we see the

world
as um frame in the theoretical
and methods-based sort of background we
have and the construct we've decided to
make
that is quite an argument but of course
this can also then improve over time
here is the picture we were able to take
of Pluto in 94
and this is the one we're able to take
in 2018 and now we're sending
robots to Mars so of course our ability
to
measure stuff improves drastically over
time so
Tycho Brahne could improve his
measurement by moving the telescope
into the basement but of course he would
not be
able to see pluto in the sort of in the
same quality as we are able to today
so measurements can continuously improve
over time
but we can only make sense of what we
see
in the concept of the theories that we
have
this is and also what Kuhn would agree
with here is a famous quantum physicist
Heisenberg who said
“what we observe is not nature itself, but
nature exposed to our method of
questioning.”
and that's sort of very much in line
with the with the argument that
corners and trying to make that we never
have this direct access to reality
we don't sort of get stimulus into our
vision and brain that it just sort of
direct image of what the world
actually looks like we shape what we pay
attention to
and what we see and how we interpret
it and how we perceive it depends
drastically on the amount of prior
training and experience that we have
so this is sort of the setup of what

Summary of Video

question was Kuhn asking
how he has done this sociological
historical approach to trying to make
sense of how does research actually work
and then trying to give a few examples
that illustrate that
researchers don't always just
immediately change their mind even if
something that
seems to be a clearly better explanation
comes along and that is because we
we can't expect all researchers just to
see sort of the same
thing out there in the world whatever we
see depends on
our prior experience and training and
our definitions and the constructs
we've decided to create and we always
sort of see the world through these
constructs through these decisions
and that is why it's difficult to figure
out if one group of researchers versus
another group of researchers is right in
the way they see the world because
well they see the world they do because
of their background
and then Kuhn is then saying there is no
absolute standard to compare that to
we'll get much more back to that in
some of the next videos and sessions

https://www.youtube.com/watch?v=CHQ7sCvwAoY

hi in this video i will present Kuhn's
concept of a paradigm so as
presented just before kuhn has this
perspective on scientific development
where normal science
does happen where science accumulates
just like building a
brick of lego towers one theory built on
top of each other
but the particular thing about his
perspective is this idea of a paradigm

change at some point
there will be challenges emerging in science that can't be solved with the existing paradigm with the existing way of
of looking at things and this is in the kind of a crisis and requires a paradigm change
so in order to understand that we need to get a good grasp of what does it this concept of a paradigm what does that actually mean so

Definition of a Paradigm

so true to put it's the fundamental assumptions that members of a scientific community
accept it's it's almost a little bit
like the organizational culture concept it's
some something implicit and taking for granted and not always reflected upon but of course something that we can decipher and figure out
um so kuhn also says it's “the entire constellations of beliefs
values and techniques shared by members of a given community”
so it's about how science is carried out
it's about what we consider
is important and what fundamental assumptions are we
building on i'll give a couple of examples in just a second
so it's both the methods and techniques that we use to collect data and analyze data but it's also some other kinds of more fundamental beliefs and also values so what do we think is important what should be
funded what's an important research question
and it's the members of a community that
share it it's how things are done in a given community if you don't

share this paradigm you're not a member
of that community you're an outsider
so again just to emphasize a paradigm
governs a group of
practitioners a group of researchers
it's not sort of a given
subject field whether it's quantum
physics or supply demand or
strategy it's a group of people that are
engaged in studying this
that share a common way of doing things
of talking about things
thinking about things which can be good
it can make collaboration
effective and communication effective
but it has some challenges and
it's it speaks to this issue of how we
see the world and how we interpret
whatever we see as i talked about in the
form of video and
i think it's useful to use this
symbolized example again about the
the mortality rate and where hand
washing helped so let's say you figure

Handwashing ex: Small Changes vs Fundamental Paradigmatic Shifts

out that
washing your hands at i don't know 30
degrees is better than 25 degrees i'm
just making this up
i don't know what is better but that
would be sort of a small change
maybe based on theoretical insight into
what a germ actually is and how we can
most effectively kill the most disease-
prone germs so that would be a small
change that would be fairly easy for us
to accept if we are told nowadays in
these COVID days that it's actually
better to wash our hands and with cold
water rather than warm water or vice
versa
it would be reasonably easy to accept
because we already accept hand washing

we already accept what a virus is
but
let's say that you were in the mid 19th
century in the 1840s
and you don't know that diseases are
infected via germs you simply don't have
any
fundamental assumption and insight and
then suddenly someone tells you that you
need to
wash your hands in order to make sure
that the mothers are about to give birth
they don't die
in in the birth bed so if you don't have
the fundamental assumption and paradigm
so to say about
what a germ is it's just it becomes
very very difficult for you to for you
to accept that hand washing works
you need the underlying insight and
premise on
what a germ is and how diseases can be
transmitted from one person to another
and so that's that way that example quite
nicely illustrates the point that smaller
changes of course we can easily adapt to
that it's not Kuhn is not
saying that any kind of change is
difficult to take in as a scientist
it's the big fundamental changes it just
completely
overthrows our usual way of seeing the
world
which makes it difficult so quite
literally again using Kuhn's perspective
the researcher that is looking at this
data is not
seeing what Semmelweis is seeing he's
not able to take in the information and
the data
he's not able to frame it in in the way
that allows him to say yes it was hand
washing that made the difference
it had to have been something else
because hand washing would be magic
it wouldn't be theory it wouldn't be
science

so that's that's Kuhn’s point here
that
paradigms refer to something very basic
that it's very difficult for
us to sort of shift away from it's just
like changing an organizational culture
is also not something you can do
from one day to another

Elements of a Paradigm

here i've tried to highlight sort of a
few more a few of the elements that
Kuhn says are part of a paradigm so just
to
to help you get a better understanding
of it
so he's highlighting these four
different elements of a paradigm in this
more detailed approach to it
so for instance we have symbolic
generalizations that supply demand curve
for instance this is just sort of a very
efficient way of communicating
how the economic system works in our
capitalistic society
or we also have sort of a fundamental
notion and paradigm about what
a human economic actor is at least in
classical economics
pre-Kahneman as we heard about the
last time about them being rational and
just following incentives we just need
to give them information
and money and then they will shift their
behavior there was a
fundamental assumption um that
that could be very difficult to change
for for a researcher
it's also about value so what
questions do we want to pursue but also
to what degree do we for instance have
to be able to predict the future as
ascientist or
can we just so to speak explain stuff
and should we do lab experiments a field
experiment or what kind of
method do we think is the most
insightful for getting to the truth but

also to
to help the world in the science that we
carry out
and what examples do we use sort of as
tecxtbooks
examples what do we think about when you
think economic theory are we
sort of focused on wall street and the
capitalistic system or
the sort of where the money flows are we
focused on the
the poor person or um what what is sort
of the
the main what are the key examples that
you would see an economic textbook
or a marketing textbook or strategy
textbook what examples are we starting
from in order to
um sort of say
these are the problems that we need to
be able to solve
so it makes quite a difference if we say
that we need to solve the problem of
poverty versus we need to
solve the problem of efficient stock
markets or efficient
capitalistic system these are two sort
of different angles of trying to go
about
and then can can impact the way science
is done to sort of follow

How Scientific Fields Change?

up a little bit on this point um so
classical economics was was is still
very much driven by for instance supply
demand curves and these
abstract insights into how economic
systems are supposed to work
the thing then is that from such a
perspective on this is what economic
theory is and this is the economic world
that we investigate
from that perspective if you increase
the supply or increase the demand well
vice versa this will have an influence
on on

on the other side i mean they're
interlinked these two things so
when in the 80s and 90s evidence began
to emerge
that said well if you increase the
minimum wage and then it
should shift the demand supply curve but
it actually turned out not necessarily
to be the case as there is quite a bit
of empirical evidence
that seems to indicate that raising the
minimum wage does not reduce
unemployment as it should
do so there was actually a noble price
winner in economics that in
the mid 80s said it's like believing
water runs uphill
of course raising the minimum wage will
have a detrimental
a bad effect on the availability of
jobs it was just
like that's literally what the quote
says it's like seeing water on uphill
it's just completely impossible
so that's also why it has probably taken
it's been difficult for
people that sort of subscribe to old
school economics to accept that raising
the minimum wage does not necessarily
lead to bad effects
because it simply goes against their
fundamental
paradigm against how the economic world
is supposed to function
and this is then also related to how
science economic science in the 16th was
carried out as you can see
was very much theory driven bit of
simulations and then you may
might have used archival data or just
data that was available from
OECD or or whatever sorurce of you
could get your hands on
very few studies where researchers went
out and collected data themselves

the red stuff here and then you look at
2011 where this data was collected
suddenly there was something like 30 35
percent of all studies
actually based on economic researchers
going out and collecting their own data
and
so we just see a shift in the way in the
methods that are used so again
this is a way of illustrating how the
fundamental ways of doing things here in
economic theory has changed quite a bit
over the last 40-50 years so this is
what kuhn is trying to get at that these
changes are
quite fundamental and and \
don't happen easily
and again sort of small changes are easy
to accept realizing that okay the
minimum wage would be
15 rather than 16 or whatever it should
be in the American states i see
uh in the U.S. as the discussion is
going on right now
but sort of implementing it versus not
implementing it is the big shift and can
be a difficult
challenge it's like the hand washing
example going from 25 to 30 degrees
could be easy
but going from no hand washing to actual
hand washing is a
substantial shift in the paradigm of the
given field
so Kuhn's point then is that yes we
often do have normal science with this
cumulative
development and we just build on
whatever research was carried out

Normal Science vs Revolutionary Science

yesterday
the paradigm is taken for granted and it
works um
we can solve problems we can improve our
explanation of how the world works

Kuhn is then pointing out that sometimes
we see large anomalies
um where there was sort of Einstein
realizing that Newton's theory of
gravity wasn't
perfect or the financial crisis in 2009
that just really highlighted that there
was something about the economic system
that we didn't quite understand yet
we can some we occasionally, rarely but
occasionally we see a crisis we there's
an anomaly there's something we simply
cannot explain in the given paradigm
that we have right now
stuff is just happening that it doesn't
make any sense um just like
Semmelweishand washing example
and this then requires a paradigm change
and this is where Kuhn isn't saying
we are not just building on top of
research from yesterday
Semmelweis was not building on
research that was done sort of the year
before
he was sort of overthrowing the
fundamental understanding
and this is why science and the large
perspective according to Kuhn is not
just always cumulative
sometimes a revolution happens and it's
important to
to realize that and understand that he
argues because it says something about
what science is and to what degree
science can be true
we'll get that in the we'll get to that
in the live session
so revolution is a reconstruction of
group commitments or group beliefs
so it's a reconstruction of the paradigm
of how things are done
let me give you a few more examples
first i want to link to this video
i won't play it here and then and this
feed here because the sound wouldn't
quite be good enough but i can recommend
going to this
video here it's just three four minutes
long and a quite

simple but useful illustration of what a
paradigm shift is and as you can see
it's yeah simple with some glasses and
some thingies that are being put into
these classes watch the video it i think
it might be useful
o so again Kuhn's point is that we
sometimes see anomalies that are so big
that we
should think well why didn't we
predict this why can't we even explain
this
should this then maybe lead to a
scientific revolution
it's not necessarily saying that this is
the case maybe we can work it out
with the existing paradigm
but he's saying that's what a crisis
sometimes can lead to when something so
fundamental happens that is so
unexplainable and so important
we might have to rethink okay is our
economic scientific framework is that
actually adequate
when something like this in 2008 can

How to Compare?

happen
but these are then also difficult to
carry out these paradigmatic
changes because
how do you compare um let's say you take
this bottom example where
we could go back to early 20th century
or late 19th
century and and say well if we wanted to
understand behavior we would sort of
look at people
that would be the fundamental assumption
if we really wanted to understand what
not just on behavior if you really want
to understand what people are thinking
and feeling
we need to look at their behavior that
was our sort of access point to
um the human insight

while nowadays we are at least to some degree able to do
various kinds of MRIsis or FMRIs or whatever they're called where you can look
inside the brain quite literally and that's of course a fundamentally different way of trying to get
data on a human being the point here is that if this researcher from let's say 2021 went back to 1921 and said look i've got a brain scan here this is what happens when we expose the
person to I don't know whatever it is we expose this person to something
and then we try to see what how their brain is responding to that i mean this would be data that would not fit the existing paradigm it would be very very difficult to
explain to this person from 1921 what was going on
you need this extensive training and education
in order to see this kind of data it wouldn't
i mean it wouldn't help wouldn't just be enough to say look at this
output look at this printout um you need sort of a lot of baggage you need a lot of education you need a lot of training in order to
make sense of scientific data and that's a key point that Kuhn is trying to make and then it's difficult to make any kind of direct comparison you
you sort of have to shift your current paradigm into a new one in order to be able to make sense of it
and so they're not straightforward to compare and we can also make this sort of a challenge of being able to compare in a different way
so

Further Barriers to Paradigmatic Change

so let's think about the the big

economic models that we have in denmark
that
try to help us predict behavior so for
instance there are economic models that
are being used
at various universities or in the danish
government
where they say well let's imagine we
changed interest rate how would that
influence and change behavior in denmark
it's important to be able to
try to predict these things in advance
because it can help you
sort of shed light on should we change
the interest rate or not this is just
an example i could have used many other
kinds of um
economic model examples so so these
people that run these models they of
course try to to to
understand and model human behavior and
the question then is how do we what
assumptions do we make what what do we
put into these models in order to be
able to predict their behavior
and and and currently the the the
general approach is still to say well
this
use this homo economicus the rational
all-knowing being
um the moment an interest rate changed
then these models assume that all danes
know about this change and they
and it just assumed that they would be
able to change their behavior already
tomorrow because they're aware of these
changes and they are they respond to
even very very small changes
so quite literally and one might sort of
begin to buy
more expensive food in the supermarket
if it's easier to borrow money and of
course
you might also buy more houses and cars
etc so that's the assumptions that is
being put into the model right now
there's another approach called an
agent-based model we don't need to sort

of know the details of that but
the point is this is a setup where
we don't necessarily need to assume that
all that everyone sort of
immediately is aware of this change in
the interest rate so the agents in this
computer model they're human beings so
to speak in this computer model they
they don't necessarily know but they
might respond to what the
neighbor does so quite literally the
point could be i didn't read the
newspaper so i didn't know the interest
rate changed
but maybe a week from now i realized
that my neighbor just bought a new car
then i asked him well why did you buy
this new car looks expensive well
then i might realize the interest rate
changed and in that way
it will sort of be a slower rollout of
awareness of
of this and and this would lead to
different predictions of how quickly and
how much a change of interest rate would
change economic behavior in denmark
very crude example but sort of
fits two different kinds of approaches
of trying to make sense of human
behavior
what assumptions are we making about the
human being but the thing is
if we were to implement the second
approach here at the governmental level
well first of all we would need new
theoretical models we need new computer
programs that have to be developed
but we also need the people that
actually sort of employed to do this at
danish government
we need them to get the different kind
of education which means that
universities also have to shift their
um their approach and then we just need
a fundamental change of practice at
governmental level and maybe it's not
enough that denmark changes maybe we
need

UN or other international
organizations to change as well
so this is sort of the sociological
institutional approach that Kuhn has on
these paradigmatic shifts it doesn't
just require one single person to try to
do something different
it can actually require a big collective
of
changing their approach and and who
changes first and there
is a complex system you need multiple
different
elements to almost change at once and
that's just not easier this could easily
take
5 10 15 20 years before we get new
employees at the government that are
able to run these different kinds of
models and these computer programs
are developed and etc right now we just
have one overall program called adam
that
relies on this assumption so so again
just
an example trying to illustrate that
there are barriers to paradigmatic
changes this is not just something that
we can easily do
and engage in and and that's Kuhn's
point that
that's cKuhn’'s point that this is not uh
this is
fundamental because it goes into the
essence and basic of stuff it's not just
changing the water temperature of hand
washing it's it's more fundamental
changes so this is the

Summary of Kuhn 1962

point here is trying to make that yes
often normal science might
go um about in the cumulative way where
we can
always build on each other and we always
get more clever and we can always say
theory from today is better than from

than one from yesterday because
it's building on the one from yesterday
it's extending it
and Kuhn s saying that sometimes we get
such a fundamental crisis
then then we need to completely rethink
the way that we approach the world and
then it's still
within sort of an overall scientific
world and framework
it's still science that we do
but it's paradigmatic shifts and he's
then trying to explain
what the paradigm is and why these
paradigmatic shifts are different difficult
and why it's difficult for a person
who doesn't believe in germ theory and
how this person
struggles with looking at hand washing
working because how and why
would it? it doesn't fit the fundamental
assumptions that the field had at the
given time
so this is what the concept of a
paradigm is trying to do
if paradigm tells researchers what is
important
what skills and and training should a
researcher have
what data will we collect what do we
teach in universities
what research will be accepted what
research will be funded
um and so coming in as an outsider and
presenting something completely
different as Semmelweis it can be
really really challenging because you're
going against the grain you're going
against a
common understanding so theory is much
much more easy to change, it's the
fundamental paradigms that are
that are argued to be difficult
and finally i should really point out that
when Kuhn uses the word paradigms he

doesn't use it in the same way as Guba
so post-positivism positivism etcetera
are not paradigms in the Kuhn’s kind of
sense so of course they're related it's
about some fundamental assumptions but
Kuhn point is about um sort of it's more
closely-related to the actual signs we
do what assumptions does a
particular researcher have in a
particular discipline um
about what methods we will be using what
what theory we believe is true how we
make sense of for instance human
behavior
etc
in the live session we will then try to
get an understanding of
trying to get more examples of
how scientific change takes place and
why paradigm shifts are difficult
but also get a more sort of general
perspective on what does this mean
for science how can how can science be
true
or how can one theory be better than
another if we are comparing two
approaches that are just fundamentally
completely different and
we don't really have a frame of
reference how can we say that a theory
is true or better
this is what we will then also be
getting to in the live session